

Example configuration for training

Create a simple neural network using SINGA's APIs:

```
import singa
from singa import tensor, opt, autograd
# Define a model
class MLP(singa.Module):
    def __init__(self):
        super().__init__()
        self.w0 = tensor.Tensor((784, 512), singa.float32)
        self.w1 = tensor.Tensor((512, 10), singa.float32)
        self.w0.gaussian(0, 0.01)
        self.w1.gaussian(0, 0.01)

    def forward(self, x):
        x = autograd.matmul(x, self.w0)
        x = autograd.relu(x)
        x = autograd.matmul(x, self.w1)
        return x

# Initialize model and optimizer
model = MLP()
sgd = opt.SGD(lr=0.01)
# Training loop
for epoch in range(10):
```

```
for x, y in dataloader: # Replace with your data loader
    x = tensor.Tensor(x)
    y = tensor.Tensor(y)
    out = model(x)
    loss = autograd.softmax_cross_entropy(out, y)
    sgd.backward_and_update(loss)
```

Apache SINGA is now configured for your machine! You can adjust the configurations based on your hardware (CPU/GPU) and use case (single-node/distributed).

Apache SINGA is ideal for developers and researchers who need scalable, efficient, and flexible deep learning with strong support for distributed environments. Its unique blend of performance optimisations, multi-modal data handling, and fault tolerance makes it a powerful alternative to mainstream frameworks like TensorFlow and PyTorch, especially in large-scale or resource-constrained scenarios. **END** 

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Cost efficiency: Automating tasks with MLOps and AIOps can significantly reduce manual effort, cutting down operational costs. You won't need to spend as much on human resources to maintain machine learning models or troubleshoot IT issues, allowing you to reinvest those savings into growth initiatives.

Scalable solutions: As your business grows, both MLOps and AIOps scale seamlessly to handle increased demand—whether it's more data for your machine learning models or larger volumes of traffic on your IT infrastructure.

Improved customer experience: With machine learning models that are constantly updated and optimised, your business can deliver more personalised and relevant experiences to customers. Plus, AIOps ensures your services are always available, reducing the risk of downtime and enhancing customer satisfaction.

Deciding whether to prioritise MLOps or AIOps depends largely on your business's primary goals and existing challenges.

If your business is focused on driving innovation through data and AI models, MLOps should be your top priority.

It's ideal for companies looking to improve model accuracy, reduce time-to-market for machine learning solutions, and ensure consistent performance.

If your organisation is more concerned with keeping IT systems running efficiently and securely—ensuring uptime, detecting issues quickly, and maintaining smooth operations—then AIOps is what you should focus on.

The key takeaway? MLOps ensures that your AI models are always working at their best, while AIOps ensures that your IT infrastructure supports these models without disruption. Together, they can drive business growth and success in an increasingly automated world. **END** 

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